School-Connect Project Brief (Revised Architecture)

1. Project Overview

School-Connect is a modern multi-tenant school management mobile application designed to digitally connect schools, parents, teachers, students, and school administration through a unified communication and operations platform. The platform will now be built using React Native with Expo and TypeScript to ensure fast cross-platform mobile development, maintainability, and scalability. Firebase services will serve as the primary backend infrastructure, enabling real-time functionality, authentication, cloud storage, push notifications, analytics, and scalable backend services without requiring traditional AWS cloud infrastructure. The system must support onboarding multiple schools under one infrastructure while allowing each school to operate within its own branded and isolated portal environment.

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2. Core Objectives

• Centralize school-parent-teacher communication
• Digitize school operations
• Provide real-time updates and notifications
• Improve transparency around fees, events, academics, and activities
• Reduce administrative workload
• Deliver an intuitive and modern user experience

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3. User Roles & Permissions

Parent Portal
• View child profiles
• Receive push notifications
• Monitor school fee payment status
• View school timetable and events
• Access lunch menus and awards sections

Teacher Portal
• Manage timetables
• Upload assignments and announcements
• Communicate with parents
• Manage sports and culture updates

School Administration Portal
• Manage students, staff, and school profiles
• Track payments and analytics
• Publish announcements and notifications
• Configure school branding and settings

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4. Multi-Tenant Architecture

The platform must support multiple schools under one infrastructure with dedicated branded portals, isolated data environments, and independent user management. Recommended structure:
• schoolname.school-connect.app
OR
• app.school-connect.app/schoolname

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5. Updated Technology Stack

Mobile Frontend: React Native with Expo
Language: TypeScript
Backend Services: Firebase
Database: Firebase Firestore
Authentication: Firebase Authentication
Storage: Firebase Storage
Push Notifications: Expo Notifications + Firebase Cloud Messaging
Analytics: Firebase Analytics
Serverless Functions: Firebase Cloud Functions

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6. Backend Infrastructure Direction

The platform will no longer use AWS infrastructure services such as EC2, ECS, EKS, RDS, S3, Lambda, or CloudFront. Firebase will provide the primary backend infrastructure and managed services for: • Authentication • Real-time database operations • Cloud storage • Push notifications • Analytics • Serverless backend functions • Hosting and scalability support This approach simplifies deployment, reduces operational overhead, accelerates development speed, and improves maintainability for the initial growth stages of the platform.

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7. Security Requirements

The application must include: • Firebase Authentication • HTTPS enforcement • Role-based access control • Multi-factor authentication support • Secure Firestore rules • Tenant-level data isolation • Audit logging support • GDPR-ready architecture

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8. Core Modules

• Real-time notification engine
• Calendar and scheduling system
• School fees and payment tracking
• Sports & culture management
• Awards & achievements section
• Lunch menu module
• Messaging and announcements

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9. UX/UI Direction

The application should maintain a clean, modern, and mobile-first interface. Design principles: • Minimal clutter • Fast navigation • Responsive layouts • Accessible interface • Maximum 3 taps to major functions

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10. Performance Requirements

• Under 2-second average response time
• 99.9% uptime target
• Support for 100+ schools
• Support for 100,000+ users
• Real-time notification delivery

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11. Development Phases

Phase 1: Authentication, portals, timetable, calendar, notifications, fees tracking.

Phase 2: Sports & culture, awards, lunch menu, messaging, online payments.

Phase 3: AI integrations, transport tracking, biometric attendance, e-learning integration.

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12. Final Direction

This platform must be engineered as a scalable education technology ecosystem capable of supporting thousands of schools and future AI-driven education services. The development team must prioritize: • Scalability • Security • Performance • Maintainability • Real-time mobile experiences • Clean architecture • Long-term extensibility The revised architecture using React Native, Expo, TypeScript, and Firebase is intended to maximize rapid development, operational simplicity, cross-platform consistency, and scalability for future AI-enabled education services.
